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WOXSEN UNIVERSITY

PROJECT DOCUMENTATION

ClaimCheck

An on-premises small language model that verifies health insurance claims, with a human making every decision

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A project by

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Small Language Model (SLM) Project

ClaimCheck

Part 0. Product Overview

ClaimCheck reads a hospital claim form and its discharge summary, pulls out 9 key fields, shows where the two documents disagree, and hands the final decision to a human reviewer. It runs on a fine-tuned 3-billion-parameter model (Qwen2.5-3B-Instruct + QLoRA) that can sit inside an insurer's or hospital's own servers, so patient data never leaves the building. ClaimCheck is a Veltris x Woxsen University collaboration.

What the product does

1. **Reads** both documents: typed text, digital PDFs, scanned PDFs and photos (OCR with Tesseract).
2. **Extracts** 9 fields from each document with the fine-tuned model, each with the exact quote it came from: patient name, age, gender, hospital, admission date, discharge date, diagnosis, procedure, total amount.
3. **Compares** the fields. Python rules check names, numbers and dates exactly. The model checks whether differently worded medical terms mean the same thing (for example "Appendectomy" vs "laparoscopic excision of the inflamed appendix").
4. **Explains** through a chat box: the reviewer can ask questions about the case, answered only from the two documents.
5. **Hands over** to a human: the reviewer clicks Approve, Reject or Send back. The AI never gives a verdict.
6. **Records** every decision in a tamper-evident audit log (each entry is chained to the previous one with a SHA-256 hash).

The problem

Indian insurers settled a record 32.6 million health claims in FY25 (Apr 2024 to Mar 2025), paying Rs 94,248 crore (IndiaMedToday, citing the IRDAI annual report). Every claim needs someone to check that the hospital's bill and the medical record tell the same story.

The time to do that is shrinking. IRDAI's May 2024 master circular asks insurers to decide cashless requests within one hour and final discharge authorisation within three hours, and makes insurers and TPAs collect documents from hospitals instead of patients (The Week).

Mistakes are expensive both ways. A BCG and Medi Assist report estimates Rs 8,000 to 10,000 crore a year leaks from fraud, waste and abuse; about 2% of claims are confirmed fraud and another 8% sit in a

grey zone (Deccan Chronicle). Wrong rejections, on the other hand, delay care and payment for genuine patients.

The market

Measure (India, FY25)	Value	Source
Health insurance premium	Rs 1,27,417 crore (up 9.19%)	Outlook Money
Health claims settled	32.6 million	IndiaMedToday
Total health claim payouts	Rs 94,248 crore	IndiaMedToday
Average payout per claim	Rs 28,910	IndiaMedToday
Claims handled by TPAs	about 69% (by number)	IndiaMedToday
Claims repudiated	about 8%	IndiaMedToday
Leakage from fraud, waste and abuse	Rs 8,000 to 10,000 crore a year	Deccan Chronicle (BCG and Medi Assist)

The FY25 claim figures come from press coverage of the IRDAI annual report; they have not been checked against the report itself.

Who it is for

User	What they do today	What ClaimCheck gives them
TPA claims processors (primary buyer)	Read every claim form against the discharge summary by hand, under the 1-hour and 3-hour limits	Fields pulled out and differences flagged in about a minute, with quotes to verify
Insurer in-house claims teams	Same check for claims not sent to a TPA (about 31%)	Same, plus an audit trail for regulators
Hospital TPA desks	Fix rejected claims after the insurer queries them	A pre-submission check that catches mismatches before the claim is sent
Audit and compliance teams	Sample past claims for errors and fraud	A tamper-evident log of every human decision

Why it must run on-premises

Claim forms and discharge summaries are personal health data, so sending them to a public AI service raises consent and data-transfer questions under India's Digital Personal Data Protection Act, 2023. A small model that runs on the insurer's own hardware avoids that: the documents, the model and the audit log all stay inside the organisation. This is also why the assignment brief rules out Claude, GPT and Gemini for the core model.

Requirements

Functional

ID	Requirement	Status
FR1	Accept claim form + discharge summary as text, PDF (digital or scanned) or image	Done
FR2	Extract the 9 fields with an exact supporting quote each	Done (fine-tuned model)
FR3	Compare names, ages, genders, dates and amounts with exact rules	Done
FR4	Judge whether differently worded diagnoses, procedures and hospital names mean the same	Built, not yet run on the real model
FR5	Answer reviewer questions about the case from the two documents only	Built, not yet run on the real model
FR6	Reviewer decides: Approve, Reject (with note) or Send back (with note)	Built and tested with a stand-in model
FR7	Log every decision in a hash-chained audit log and show if it was tampered with	Built and tested
FR8	Pull claims automatically from an insurer's claims system	Planned (only if sold)

Non-functional

ID	Requirement	How ClaimCheck meets it
NFR1 Privacy	No document leaves the organisation	Model runs locally; no external API calls
NFR2 No invented numbers	Amounts, dates and ages are never judged by the model	Python rules only for these fields
NFR3 Explainability	Every extracted value is traceable	Each field carries its exact quote; each meaning check gives a one-line reason
NFR4 Human control	The AI cannot approve or reject	Findings only; decision buttons are human-only
NFR5 Auditability	Decisions cannot be edited silently	SHA-256 hash chain, verified on every load
NFR6 Low-cost hardware	Runs without a large GPU cluster	4-bit model fits a free 16 GB T4 GPU; a CPU-only build is possible
NFR7 Speed	A case reviewed within the 1-hour cashless window	Target: under 2 minutes per case; to be measured

Tech stack

Layer	Tool	Why
Base model	Qwen2.5-3B-Instruct	Strong instruction following at 3B; runs on a free T4
Fine-tuning	QLoRA: PEFT (LoRA r=8, alpha=16) + bitsandbytes 4-bit NF4	Trains about 0.4% of weights on a free GPU
Training	Hugging Face Transformers, TRL SFTTrainer	Standard supervised fine-tuning loop
OCR	Tesseract (pytesseract), PyMuPDF	Free, local; PyMuPDF reads digital PDFs and renders scans for OCR
Comparison	Python rules (regex, date parsing) + model meaning check	Exact where it must be, flexible where wording varies
User interface	Streamlit (earlier demo: Gradio)	Upload, findings table, chat and decision screen in one Python app
Audit log	JSON Lines file + SHA-256 hash chain	Simple, tamper-evident, no database needed
Compute	Google Colab and Kaggle (free NVIDIA T4, 16 GB)	Training and demo hosting at zero cost
Public link	Cloudflare quick tunnel	Shares the running app without a server
Data	Synthetic generator (seed 42): 1,500 / 150 / 400 cases	No real patient data used anywhere

Part 1. Project Status Report

ClaimCheck is trained, evaluated and working end to end in a demo app: the fine-tuned model reached 97.5% field accuracy on the outputs it formatted correctly, and the first real test case was read, compared and approved correctly. Two things are open: a permanent public link, and running the new Streamlit app (chat and meaning check) on the real model.

Project: ClaimCheck, on-premises claim verification with a small language model **Collaboration:** Veltris (Vertical AI Academy) x Woxsen University, SLM course assignment **Author:** Shaik Shazia Mahek (23WU0102185), B.Tech CSE (AI & ML), Woxsen University

Version history

Version	When	What changed
v1.3	14 Oct 2026	Streamlit app: findings table, model meaning check, chat, human Approve / Reject / Send back, review-complete screen. 3 demo cases as realistic clean and scanned PDFs
v1.2	14 Oct 2026	Pipeline bugs fixed (wrong Male/Female match, prompt mismatch with training, prompt truncation). Moved to Kaggle; first end-to-end test passed
v1.1	Oct 2026	QLoRA training (1 epoch, about 188 steps) and evaluation on 100 test samples done; Gradio demo built
v1.0	Oct 2026	Architecture fixed; synthetic dataset generated; base model switched from Qwen3-4B to Qwen2.5-3B (see ADR-01)
v0.x	From the first week of Veltris	Earlier SLM experiments (Genshin lore, from-scratch model) dropped for a real industry use case

Work status

Workstream	Status	Notes
Synthetic dataset	Done	1,500 train / 150 val / 400 test cases; 30% with deliberate mismatches; 5 OCR noise levels
Fine-tuning	Done	Adapter saved (r=8, alpha=16, 7 target modules)
Evaluation	Done	97.5% field accuracy, 70% valid JSON (Part 4)
Pipeline (OCR, extract, compare, audit)	Done	Tested on the matching test pair: 17 of 17 values correct, verdict as expected
Streamlit app with chat and human decision	Built	Tested with a stand-in model; real-model run pending
Demo documents	Done	3 cases x 2 documents, clean and scanned; OCR reads every key value
Always-on public hosting	Open	Colab link works only while the notebook runs (Part 2, alternatives)
Model comparison (Phi-3-mini, Gemma-2-2B, Qwen2.5-1.5B)	Not started	Needs evaluation runs on the same 400-case test set
GitHub repository and README	Not started	
Marketing video	Done	86-second promo with AI voiceover, music and subtitles; app answers in the demo shots are scripted

Risks and blockers

- **30% of outputs are not valid JSON.** A repair step recovers some; the rest show as "extraction failed" and need a retry or manual entry.
- **Free compute ends sessions.** Colab disconnects after inactivity, so the public link is not permanent.
- **Meaning check and chat are untested on the real model.** Their quality depends on the base Qwen model, not the fine-tune.
- **Synthetic data only.** Accuracy on real hospital documents is unknown until a pilot with real (consented) data.

Next steps

- [ ] Run the Streamlit app on Colab with the real model; test all 3 demo cases
- [ ] Record screenshots of each screen for this document
- [ ] Choose permanent hosting (Part 2, alternatives)
- [ ] Evaluate the comparison models on the same test set
- [ ] Push code, notebooks, adapter and test documents to GitHub
- [ ] Produce the marketing video (done 15 Oct 2026; swap in a real-model recording if required)

Open question: names of the Veltris mentor and the Woxsen faculty guide, and the submission date, for the cover.

Part 2. Design Doc

The design splits the work by risk: the fine-tuned model reads, exact Python rules judge anything numeric, the model judges only wording, and a human makes every decision. Everything runs inside the organisation.

Context and scope

A health claim arrives with a claim form (filled by the insured, with the bill) and a discharge summary (written by the hospital). A processor must confirm both describe the same patient, stay, diagnosis, procedure and amount. ClaimCheck covers this cross-check for one claim form and one discharge summary per case. It does not read itemised bills, lab reports or policy wordings.

Goals and non-goals

Goals

- Pull the 9 fields from both documents, including scans and photos, with a quote for each
- Flag every difference a processor would need to look at, and explain it in one line
- Let the reviewer ask questions about the case
- Keep a tamper-evident record of who decided what
- Run on hardware an Indian insurer or TPA already has, with no external AI service

Non-goals

- Approving or rejecting claims automatically
- Checking whether a treatment is covered by the policy
- Detecting fraud patterns across many claims
- Medical coding (ICD-10 assignment) or bill auditing

Architecture

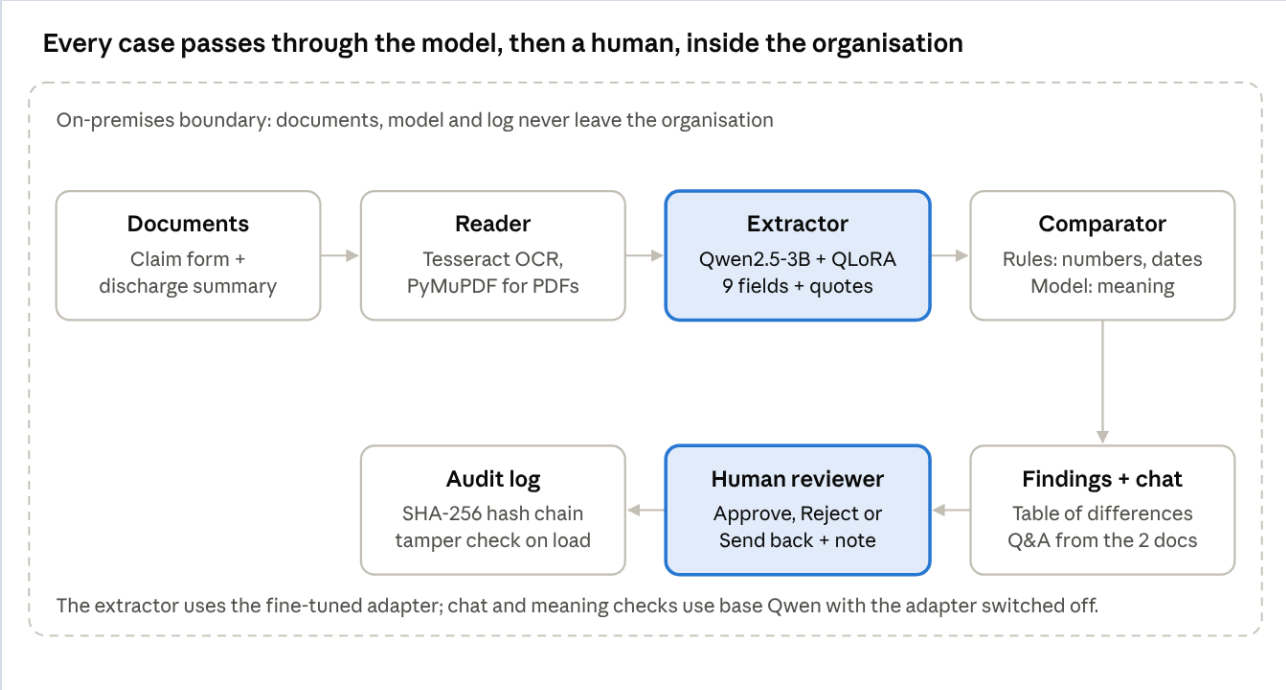


Figure 1. ClaimCheck architecture: 7 components, 1 human decision

A case flows left to right through reading, extraction and comparison, then back through the findings screen to the reviewer, whose decision is logged.

How each field is checked

Field	Checked by	Rule
Patient name	Python	Same words after removing titles (Mr, Smt, Dr); a one-letter typo is a difference
Age	Python	Same number ("52 years" = "52")
Gender	Python	Same first letter (F = Female); "male" never matches "female"
Admission and discharge dates	Python	Parsed as dates in 11 Indian and ISO formats, then compared
Total amount	Python	Parsed as a number (Rs, commas, "/-" removed); claim-only field
Diagnosis, procedure, hospital	Python, then the model	Same words first; if not, the model answers SAME or DIFFERENT with a reason

Each extracted field is stored as `{"value": ..., "quote": ...}`, so a reviewer can see exactly which line of the document a value came from.

Alternatives considered

Option	Decision	Why
Keyword matcher instead of a model	Rejected	Baseline on the messy test set: 90.5% field accuracy but 12.5% of cases wrongly agreed; cannot read a narrative paragraph
Cloud AI (Claude, GPT, Gemini)	Rejected	Sends health data outside the organisation; ruled out by the assignment brief
Qwen3-4B as base model	Replaced	Kaggle's built-in transformers 4.47.1 cannot load the Qwen3 architecture (ADR-01)
Model decides the verdict	Rejected	A wrong automatic rejection harms a patient; responsibility must sit with a person (ADR-04)
Model compares numbers and dates	Rejected	A small model can misread digits; rules are exact (ADR-03)
Retraining for the comparison task	Not needed	The extractor already turns narrative text into clean values; base Qwen handles synonyms
Hugging Face Spaces (free)	Not available	New free accounts cannot host Gradio apps on free CPU since mid-2026
Streamlit Community Cloud	Rejected	2.7 GB memory limit is too close to the model's size

Option	Decision	Why
Vercel, Render free tier	Rejected	Built for web apps; free memory far below what a 3B model needs
Oracle Cloud Always Free (4 CPU, 24 GB)	Shortlisted	Always on and free, needs a CPU (GGUF) build of the model; signup capacity is uncertain
Colab + Cloudflare tunnel	Chosen for now	Works today on a free GPU; link lasts only while the notebook runs

Cross-cutting concerns

- **Privacy and security:** no external API calls; the model, documents and log stay on the organisation's machines, in line with the DPDP Act 2023. Demo data is fully synthetic.
- **Reliability:** if the model's output is not valid JSON, a repair step tries to fix it; if that fails, the screen says extraction failed and shows the raw output instead of guessing.
- **Human control:** the AI shows findings only; Reject and Send back require a written note.
- **Auditability:** every decision is chained to the previous one with SHA-256; the app re-checks the whole chain each time it loads and warns if any entry was edited.
- **Observability:** the processing log shows how each document was read (text, PDF or OCR), how many characters, and how many fields were found.
- **Cost:** training and the demo run on free T4 GPUs; no paid service is required.

Part 3. Architecture Decision Records

Ten decisions shape ClaimCheck; each record below gives the situation at the time, what was chosen, and what that choice costs. Format: Michael Nygard's ADR template (status, context, decision, consequences).

ADR	Decision	Status
ADR-01	Qwen2.5-3B-Instruct as the base model	Accepted (replaces Qwen3-4B)
ADR-02	Fine-tune with QLoRA (4-bit, LoRA r=8)	Accepted
ADR-03	Python rules judge numbers, dates, names; the model judges only wording	Accepted

ADR	Decision	Status
ADR-04	The AI shows findings; a human makes every decision	Accepted
ADR-05	Synthetic data only	Accepted
ADR-06	Train with fp16 and bf16 both off	Accepted
ADR-07	One model: adapter on to extract, off to chat and judge meaning	Accepted
ADR-08	No external AI service; built to run on-premises	Accepted
ADR-09	Audit log as a SHA-256 hash chain in a JSON Lines file	Accepted
ADR-10	Demo hosting on Colab + Streamlit + Cloudflare tunnel	Accepted (temporary)

ADR-01. Qwen2.5-3B-Instruct as the base model

Context: The first choice was Qwen3-4B-Instruct. Kaggle's built-in transformers 4.47.1 cannot load the Qwen3 architecture, and the free T4 GPU has 16 GB. **Decision:** Use Qwen2.5-3B-Instruct. **Consequences:** Loads on standard free notebooks and fits in 4-bit with room to train. Slightly smaller model, so some reasoning headroom is lost; the comparison against Phi-3-mini, Gemma-2-2B and Qwen2.5-1.5B (Part 4) will test whether the choice holds.

ADR-02. Fine-tune with QLoRA

Context: Full fine-tuning of 3 billion weights needs far more than 16 GB of GPU memory. **Decision:** Load the base model in 4-bit NF4 with double quantisation and train LoRA adapters (rank 8, alpha 16, dropout 0.05) on all 7 attention and MLP projections. **Consequences:** About 12 million trainable weights (about 0.4%); the adapter is a small separate file that can be swapped or versioned. 4-bit loading needs an NVIDIA GPU; a CPU build needs the adapter merged and converted (GGUF).

ADR-03. Rules for numbers, the model for wording

Context: A small model can misread or invent digits, and a wrong amount or date is the costliest error in claims. Medical wording, on the other hand, varies too much for rules. **Decision:** Names, age, gender, dates and amounts are compared only by Python rules. Diagnosis, procedure and hospital name go to the model only when the rules find a difference. **Consequences:** No numeric verdict can be hallucinated. Wording checks depend on base Qwen's medical knowledge and may be marked "unclear".

ADR-04. Findings, not verdicts

Context: An early pipeline produced APPROVED / REJECTED automatically. A wrong automatic rejection delays a patient's care or payment, and regulators expect a named person to be accountable. **Decision:** The AI shows matches, differences and reasons. Only the reviewer can Approve, Reject or Send back; Reject and Send back need a note. **Consequences:** Slower than full automation, but every decision has an owner, a reason and a log entry.

ADR-05. Synthetic data only

Context: Real claim documents are personal health data and are not lawfully available for a student project. **Decision:** Generate all training and test data in-notebook (seed 42): 1,500 / 150 / 400 cases, 30% with deliberate mismatches (diagnosis swap, amount inflation, name typo, date off by one), 5 OCR noise levels, 12% long records. **Consequences:** Reproducible and privacy-safe. Real-world accuracy is unknown until a consented pilot.

ADR-06. Mixed precision off during training

Context: Training crashed with `NotImplementedError: "_amp_foreach_non_finite_check_and_unscale_cuda" not implemented for 'BFloat16'`. The 4-bit Qwen layers produce bfloat16 values at run time, and the fp16 gradient scaler has no bfloat16 kernel. **Decision:** Three fixes together: set the model config dtype to float16 after loading; cast any remaining bf16 weights to fp16 (and LoRA weights to fp32) after preparing for k-bit training; set `fp16=False, bf16=False` so no gradient scaler is created. **Consequences:** Training is stable on the T4. It runs somewhat slower than mixed precision.

ADR-07. One model, two modes

Context: The fine-tune taught extraction and strict JSON output. Using it for conversation risks JSON-shaped or stiff answers; loading a second model doubles memory. **Decision:** Keep one base model in memory. Extraction runs with the LoRA adapter on; chat and meaning checks run with it switched off (`disable_adapter()`). **Consequences:** No retraining and no extra memory. Chat quality is that of base Qwen2.5-3B: good for questions about the two documents, limited for deep reasoning.

ADR-08. No external AI service

Context: Health data falls under the DPDP Act 2023; the assignment rules out Claude, GPT and Gemini. **Decision:** Every model call happens in the same process as the app. No document text is sent to any outside API. **Consequences:** Deployable inside an insurer's network. The organisation must supply its own compute.

ADR-09. Hash-chained audit log

Context: Reviewers' decisions must be provable later, and a database is extra setup for a demo. **Decision:** Append each decision to a JSON Lines file; each entry stores the previous entry's hash and its own SHA-256 hash. The app re-verifies the chain on every load. **Consequences:** Any edit to a past entry is detected. A file can still be deleted whole, so production needs write-once storage or a database with the same chaining.

ADR-10. Demo hosting on Colab

Context: Free always-on hosts either cannot hold a 3B model (Vercel, Render free, Streamlit Cloud) or now need a paid plan (Hugging Face Spaces for Gradio). **Decision:** Run the Streamlit app on a free Colab T4 and expose it with a Cloudflare quick tunnel. **Consequences:** Works today at zero cost. The link changes each run and stops when the notebook stops. Revisit with Oracle Cloud Always Free plus a CPU (GGUF) build.

Part 4. Model Card

The ClaimCheck extractor gets 97.5% of fields right when it produces valid JSON, but only 70% of its outputs are valid JSON, so output formatting, not reading, is the main weakness. Structure follows the Hugging Face model card.

Model details

Item	Value
Name	ClaimCheck extractor (claimcheck-qwen2.5-3b-lora-adapter)
Base model	Qwen/Qwen2.5-3B-Instruct
Method	QLoRA: 4-bit NF4 base, double quantisation, LoRA adapters
LoRA settings	r=8, alpha=16, dropout 0.05; q, k, v, o, gate, up, down projections
Trainable weights	about 12 million of about 3 billion (about 0.4%)
Adapter size	59.9 MB (31.8 MB zipped)
Output	JSON with value + exact quote for 9 fields
Author	Shaik Shazia Mahek, Woxsen University (Veltris x Woxsen)

Intended use

- Extracting the 9 claim fields from Indian hospital claim forms and discharge summaries, in English, as input to a human review
- Running inside an insurer, TPA or hospital network, on its own hardware

Out of scope: approving or rejecting claims; any language other than English; documents other than claim forms and discharge summaries (bills, lab reports, prescriptions); clinical advice to patients.

Training

Item	Value
Data	1,500 synthetic cases (3,000 documents); 150 validation cases
Noise	5 OCR levels: clean, light, moderate, heavy, terrible; 12% long records (60 to 120 lines)
Trainer	TRL SFTTrainer, Qwen chat template, same system prompt as at inference
Schedule	1 epoch, about 188 steps; learning rate 2e-4, cosine
Batch	2 per device x 8 gradient accumulation = 16
Optimiser	paged_adamw_8bit, gradient checkpointing on
Precision	fp16 and bf16 off (ADR-06)
Hardware	1 x NVIDIA T4 16 GB (Google Colab, free)
Training time and emissions	Not recorded

Evaluation

Evaluated on 100 of the 400 held-out test cases. 70 outputs parsed as valid JSON; field accuracy on those was 97.5% overall.

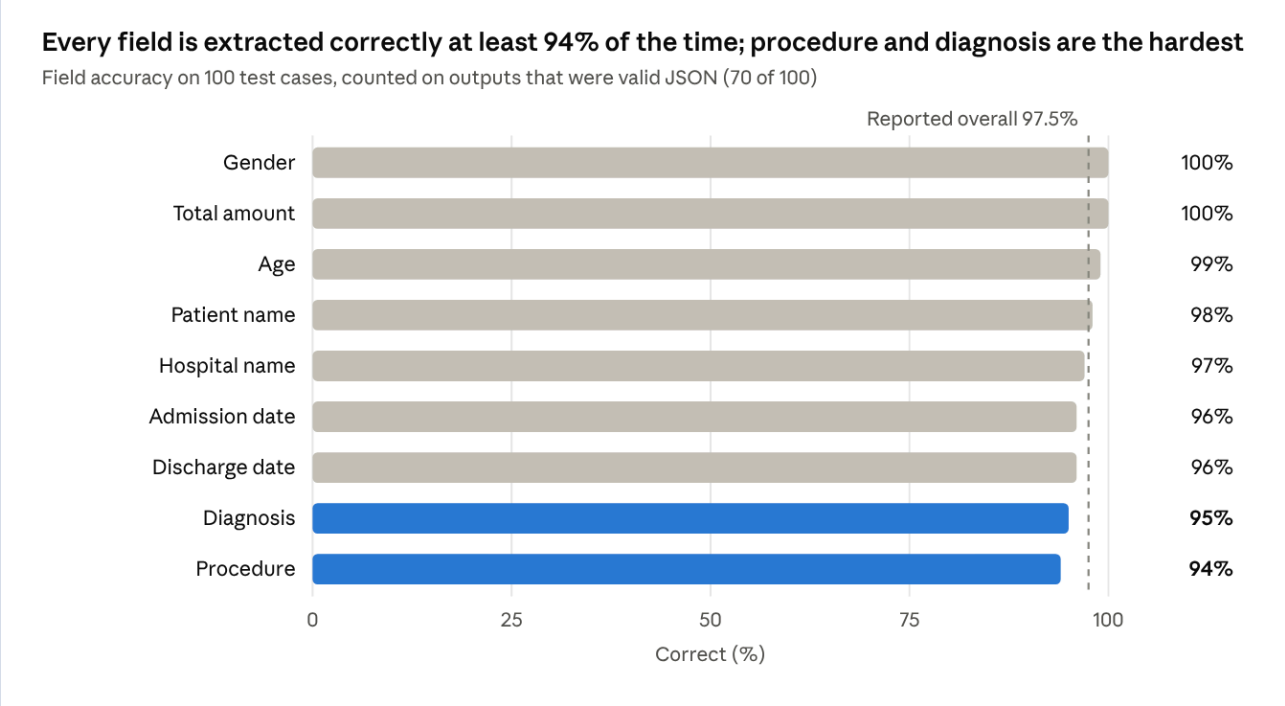


Figure 2. Field accuracy on 100 test cases (valid-JSON outputs)

The two weakest fields are the ones written most freely in discharge summaries. A live end-to-end test on the matching demo pair (14 Oct 2026) extracted all 17 values correctly and correctly returned null for the amount missing from the discharge summary.

Planned comparison

Model	Parameters	Status
Qwen2.5-3B-Instruct + QLoRA (ClaimCheck)	about 3B	Evaluated (above)
Phi-3-mini-4k-instruct	about 3.8B	Not yet run
Gemma-2-2B-it	about 2.6B	Not yet run
Qwen2.5-1.5B-Instruct	about 1.5B	Not yet run

Each will be fine-tuned with the same data and settings and scored on the same test set, so the differences come from the model alone.

Limitations and risks

- **Format failures:** 30% of outputs need JSON repair or fail; the app shows the failure instead of guessing.
- **Synthetic-only training:** real hospital layouts, handwriting and regional abbreviations may lower accuracy.

- **OCR dependence:** very poor scans can lose words before the model sees them.
- **Short evaluation:** 100 cases, 1 epoch; numbers may move with a larger test or longer training.
- **Mitigations:** every value carries its quote for checking; numbers are compared by rules (ADR-03); a human decides every case (ADR-04).

How to load it

```
from transformers import AutoModelForCausalLM, AutoTokenizer, BitsAndBytesConfig
from peft import PeftModel
import torch

bnb = BitsAndBytesConfig(load_in_4bit=True, bnb_4bit_quant_type="nf4",
                         bnb_4bit_compute_dtype=torch.float16,
bnb_4bit_use_double_quant=True)
tokenizer = AutoTokenizer.from_pretrained("./claimcheck-adapter")
base = AutoModelForCausalLM.from_pretrained("Qwen/Qwen2.5-3B-Instruct",
                                            quantization_config=bnb,
device_map="auto")
model = PeftModel.from_pretrained(base, "./claimcheck-adapter").eval()
```

Part 5. Runbook

The app starts from one Colab notebook (`claimcheck_streamlit_colab.ipynb`) in about 5 to 10 minutes and gives a public link that works while the notebook runs.

Start the app

1. Open the notebook in Google Colab. Choose **Runtime > Change runtime type > T4 GPU**.
2. Choose **Runtime > Run all**.
3. When asked, upload `claimcheck-qwen2.5-3b-lora-adapter.zip`. The notebook unzips it and downloads Qwen2.5-3B-Instruct.
4. Wait for Step 5 to print `ClaimCheck is live at: https://....trycloudflare.com`.
5. Open the link. The first visit takes 1 to 2 minutes while the model loads; later visits are immediate.
6. Keep the Colab tab open. Free Colab disconnects after a period of inactivity; rerun Step 5 for a new link.

Review a case

1. Upload the claim form and discharge summary (or paste their text) and click **Analyse claim**.
2. Read the findings table: Match, Same meaning, Different, Missing or Unclear, each with a note.
3. Open the original documents or ask questions in the chat box if anything is unclear.
4. Enter your name, add a note if rejecting or sending back, and click the decision.
5. The review-complete screen shows the case number and audit hash; click **Start next review**.

Troubleshooting

Symptom	Likely cause	Fix
"No GPU" error in Step 1	Runtime is CPU	Change runtime type to T4 GPU and run all again
Step 5 prints no link	Cloudflare tunnel slow to start	Rerun Step 5; check the tunnel log it prints
Page loads but shows an error	App crashed on start	Run Step 6 to see the app log
"Extraction failed" with raw output	Model output was not valid JSON	Click Analyse again, or paste the text instead of a scan
Link stopped working	Colab session ended	Rerun the notebook; a new link is created
A scanned field reads wrongly	Poor scan quality	Use the clean PDF or paste the text; check the quote shown

Appendix

A. Demo documents

Three fictional cases, each as a clean PDF and a scanned (image-only) PDF; OCR reads every key value in all three scanned sets.

Case	Patient	What differs	Expected result
1. Matching	Meena Patel, 52 F, gallstones	Nothing	All fields agree; reviewer can approve
2. Mismatch	Rajesh Kumar Sharma, 45 M, heart	Diagnosis (STEMI vs unstable angina); discharge date 14/09 vs 15/09; procedure worded differently	2 differences flagged; procedure marked same meaning

Case	Patient	What differs	Expected result
3. Same meaning	Farhan Ali Khan, 24 M, appendix	Procedure: "Appendectomy" vs "Laparoscopic excision of the inflamed vermiform appendix"	Procedure marked same meaning; all fields agree

All people, hospitals, the insurer (Sahayog General Insurance), policy numbers and phone numbers are made up, and each page carries a synthetic-specimen footer.

B. Glossary

Term	Meaning
SLM	Small language model: a model of a few billion weights that can run on one GPU or a CPU
LoRA	Low-Rank Adaptation: trains small add-on matrices instead of the whole model
QLoRA	LoRA on a model loaded in 4-bit, so it fits on a small GPU
NF4	4-bit NormalFloat, the number format used to store the base model
Adapter	The small file of trained LoRA weights that sits on top of the base model
OCR	Optical character recognition: turning a scanned image into text
TPA	Third-party administrator: a company that processes claims for insurers
Cashless claim	The insurer pays the hospital directly instead of reimbursing the patient
Discharge summary	The hospital's record of a stay: diagnosis, procedure, course, medicines
ADR	Architecture Decision Record: one page per design decision
Hash chain	Each log entry contains the previous entry's hash, so editing one breaks the chain
DPDP Act	India's Digital Personal Data Protection Act, 2023

C. Sources

- IndiaMedToday: record health insurance claims settled in FY25
- Outlook Money: IRDAI 2024-25 annual report, health premiums
- The Week: IRDAI master circular on health claims (May 2024)
- Deccan Chronicle: BCG and Medi Assist report on fraud, waste and abuse
- Hugging Face: Spaces overview (free hardware and plan rules)
- Streamlit Community Cloud: resource limits

- Design Docs at Google
- Michael Nygard decision record template
- Hugging Face annotated model card